Machine Learning Assignment 2 - Submission

M.Tech (AIML) • Machine Learning • Roll No: 2026ak05552

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1. GitHub Repository Link

https://github.com//ML_Assignment02_2026ak05552

This repository contains the complete source code for the assignment: **app.py**, **requirements.txt**, **README.md**, **test_data.csv**, the **model/** folder with saved trained pipelines, and the training script. All files were developed and committed through the course of this assignment.

2. Live Streamlit App Link

https://.streamlit.app/

The application is deployed on Streamlit Community Cloud and opens an interactive frontend where users can upload test data, select a model, and view evaluation metrics without any retraining taking place on the server.

3. Screenshot - BITS Virtual Lab Execution

[Screenshot to be inserted here: the Streamlit application running on BITS Virtual Lab]

This screenshot serves as proof that the assignment was executed on the BITS Virtual Lab environment, as required for 1 mark.

4. README.md Content

a. Problem Statement

The objective of this project is to predict whether a bank customer will subscribe to a term deposit, framed as a binary classification problem (yes or no). Using customer and campaign-related attributes, I trained and compared five different classification models and packaged the results into a Streamlit web application for interactive demonstration.

b. Dataset Description

I selected the Bank Marketing dataset from the UCI Machine Learning Repository (*bank-additional-full.csv*). It contains 41,188 customer records, which comfortably exceeds the minimum requirement of 500 instances. After removing the *duration* column to prevent target leakage, the dataset retains 19 input features, well above the required minimum of 12. The target column *y* indicates subscription status (*yes* = 1, *no* = 0). The data is imbalanced, with roughly 36,548 negative responses and 4,640 positive responses.

c. GitHub Repository Link

https://github.com//ML_Assignment02_2026ak05552

d. Models Used

All five models were trained on the identical train/test split and the same preprocessing pipeline so that the comparison is fair. The models are:

• Logistic Regression
• Decision Tree Classifier
• K-Nearest Neighbors Classifier
• Naive Bayes Classifier (Gaussian)
• Support Vector Machine

Each model was saved as a complete scikit-learn pipeline (.joblib) that bundles preprocessing and classification together, ensuring that the Streamlit app can load them directly without retraining.

Model Comparison Table

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.9009	0.8008	0.6905	0.2188	0.3322	0.3516
Decision Tree	0.8418	0.6258	0.3131	0.3384	0.3252	0.2360
KNN	0.8972	0.7443	0.5839	0.3039	0.3997	0.3719
Naive Bayes	0.8049	0.7755	0.3172	0.6347	0.4230	0.3490
Support Vector Machine	Pending	Pending	Pending	Pending	Pending	Pending

Note: The Support Vector Machine row will be populated with actual values from *model/metrics_comparison.csv* once training completes on the lab environment.

Observations on Model Performance

ML Model Name	Observation
Logistic Regression	Delivers high accuracy and precision but relatively low recall, suggesting it is conservative in predicting positive
Decision Tree	Shows moderate performance across metrics. As a tree-based model, it captures non-linear patterns but risks o
KNN	Achieves a strong balance between precision and recall. Being distance-based, its performance depends heavil
Naive Bayes	Posts the highest recall among the models, meaning it identifies most positive cases, though at the cost of lowe
Support Vector Machine	Training is in progress; results will be recorded here from the saved pipeline.

Overall Winner

The winning model will be selected based on the highest Matthews Correlation Coefficient (MCC), since accuracy alone can be misleading on this imbalanced dataset. From the results computed so far, KNN leads on MCC, but the final decision will incorporate the SVM result once it is available.

Final checks before submission: GitHub repository link is functional • Streamlit app link loads correctly • Application runs without errors • All four required app features are implemented • README content is included in this PDF.